FERRAN HERNANDEZ CARALT

PhD Researcher, University of Cambridge | Graph ML · Geometric Deep Learning · Higher-Order Neural Networks
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RESEARCH INTERESTS

Foundations and limitations of graph neural networks; long-range information propagation and over-squashing; sheaf and topological neural networks; principled benchmarking and evaluation of neural architectures.

EDUCATION

University of Cambridge, PhD Candidate *Cambridge, UK | Oct 2025 – Oct 2028 (expected)*
Supervisor: Prof. Pietro Liò | Fully funded by a La Caixa Postgraduate Fellowship

- Researching the foundations and limitations of graph neural networks and their higher-order variants, with a focus on long-range information propagation and over-squashing.
- Showed that commonly used benchmarks for long-range graph tasks do not reliably measure long-range capability. Introduced **TRIP** and **GRIP**, synthetic frameworks that let us provably and quantifiably test it.
- Led this work to a **first-author NeurIPS 2026 paper** (Evaluations & Datasets Track) [link]
- Collaborate with researchers at Queen Mary University of London, Microsoft, and other departments at Cambridge.

Universitat Politècnica de Catalunya (UPC) — CFIS *Barcelona, Spain | Sep 2019 – May 2024*
BSc Mathematics (GPA 8.8/10) & BSc Data Science and Engineering (GPA 9.0/10), double degree
Admitted to CFIS (Interdisciplinary Higher Education Center) [link], a selective programme that takes about 40 students per year from across Spain. Graduated with Honors in Numerical Calculus.

SELECTED PUBLICATIONS

Conference

- **F. Hernandez Caralt**, S. Heilig, A. Bazaga, A. Fischer, M. Eliasof, P. Liò. *Get a GRIP, this will be a long TRIP: A Quantifiable Long-Range Framework for Verifying Over-squashing.* NeurIPS 2026, Evaluations & Datasets Track. **First author.** [pdf]
- B. Zaz, M. Gonzàlez i Català, **F. Hernandez Caralt**, M. Eliasof, P. Liò. *Graph Transductive Sharpening: Leveraging Unlabeled Predictions in Node Classification.* Learning on Graphs Conference (LoG) 2026. [pdf]

Workshop

- M. Niedostatek*, **F. Hernandez Caralt***, R. Wang, F. Baccini, L. Giambagli, P. Liò, G. Bianconi. *Cup and Cap Topological Neural Network.* NeurReps Workshop, NeurIPS 2026 (PMLR). **Equal contribution.** [pdf]
- **F. Hernandez Caralt**, M. Gonzàlez i Català, A. Bazaga, P. Liò. *On the Necessity of Learnable Sheaf Laplacians.* GRaM Workshop, ICLR 2026. **First author.** [pdf]
- **F. Hernandez Caralt**, G. Bernárdez Gil, I. Duta, P. Liò, E. Alarcón Cot. *Joint Diffusion Processes as an Inductive Bias in Sheaf Neural Networks.* GRaM Workshop, ICML 2024 (PMLR). **First author.** [pdf]

** equal contribution*

RESEARCH & INDUSTRY EXPERIENCE

University of Cambridge, Undergraduate Researcher (Bachelor's thesis) *Cambridge, UK | Sep 2023 – Feb 2024*

- Studied opinion dynamics as an inductive bias in sheaf neural networks with Prof. Pietro Liò, leading to a first-author ICML GRaM 2024 paper [pdf].
- Presented the work in a seminar at the University of Cambridge (talk).

Axiomatic AI, Junior AI Engineer *Barcelona, Spain | Oct 2024 – Sep 2025*

- Developed an agentic AI system that automates the design of photonic integrated circuits, taking a natural-language request through to a fully optimized circuit.
- Contributed to the development and testing of the system, which was supported by a €1.6M grant [link].

Qualcomm, Research Engineering Intern *Barcelona, Spain | Jun 2024 – Sep 2024*

- Worked on R&D and MLOps for parallelizing the training and inference of large language models.
- Implemented and benchmarked Megatron-LM against DeepSpeed for LLM training and inference on clusters of up to 64 GPUs.

QuantumBlack, McKinsey & Company, Data Science Intern *Madrid, Spain | Jul 2023*

- Built a SHAP-based model explainability module in Python and integrated it into an existing Keras pipeline.

CFIS, UPC, Teaching Assistant *Barcelona, Spain | Sep 2022 – Jun 2023*

- Awarded a one-year teaching assistantship; ran seminars on machine learning, commutative algebra, and induction for first-year double-degree students.

Hewlett Packard (HP), Machine Learning Intern *Barcelona, Spain | Jul 2021 – Sep 2021*

- Developed clustering and convolutional autoencoder methods (TensorFlow, C++) to compress thermal images for 3D printers, cutting storage by 75% with under 1% error.

AWARDS

La Caixa Postgraduate Fellowship Abroad [link] *Oct 2025 – Oct 2027*
Competitive fellowship that fully funds the first two years of my PhD at Cambridge.

CFIS Scholarship, UPC [link] *Sep 2019 – Feb 2024*
Covered tuition and monthly accommodation for the double degree. CFIS selects only 40 students per year in Spain.

TECHNICAL SKILLS

Programming: Python, C++, SQL, R, MATLAB
Machine learning: PyTorch, Transformers, TensorFlow/Keras, scikit-learn, NumPy, pandas, PySpark, OpenCV
Distributed ML: Megatron-LM, DeepSpeed
Tools: Git/GitHub, Linux (Ubuntu), Conda
Languages: Catalan and Spanish (native), English (C2), German (basic)